Lab 8.1: OOP in F# (Classes, Inheritance, Polymorphism)

lab8.1.fsx

```
open System

type Shape(name: string) =
    member _.Name = name
    abstract Area : unit -> float
    default _.Area() = 0.0

type Circle(radius: float) =
    inherit Shape("Circle")
    override _.Area() = Math.PI * radius * radius

type Rectangle(width: float, height: float) =
    inherit Shape("Rectangle")
    override _.Area() = width * height

type Triangle(baseLength: float, height: float) =
    inherit Shape("Triangle")
    override _.Area() = 0.5 * baseLength * height

let circle = Circle(5.0)
let rectangle = Rectangle(4.0, 6.0)
let triangle = Triangle(3.0, 4.0)

printfn "=== Individual objects ==="
printfn "%s area = %.2f" circle.Name (circle.Area())
printfn "%s area = %.2f" rectangle.Name (rectangle.Area())
printfn "%s area = %.2f" triangle.Name (triangle.Area())

printfn "\n=== Polymorphism: one list of Shape ==="
let shapes : Shape list = [ circle :> Shape; rectangle :> Shape; triangle :> Shape ]
shapes |> List.iter (fun s -> printfn "%-10s -> Area() = %.2f" s.Name (s.Area()))

let totalArea = shapes |> List.sumBy (fun s -> s.Area())
printfn "\nTotal area of all shapes = %.2f" totalArea
```